

Cardano Incentives



Enabling a Fair Decentralized System

About myself



- PhD in Pure Mathematics from Regensburg University (Germany).
- Postdoc at Cambridge University (UK).
- Ten years working in Software Development prior to joining IOG.
- Haskell enthusiast for more than 15 years.
- Joined IOG November 2016.
- Director of Education at IOHK.
- Led the "Incentives" workstream.

Doing the Hard Work



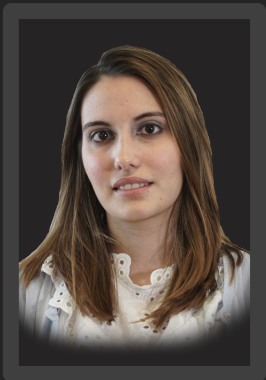
Prof. Dr. Aggelos Kiayias

Chief Scientist



Prof. Dr. Elias Koutsoupas

Senior Research Fellow



Aikaterini-Panagiota Stouka

Researcher

What are Incentives?



- **Incentives** in the context of a cryptocurrency are ways of encouraging people to participate in the protocol and to follow it faithfully.
- In the case of Cardano:
 - Being online and creating a block when having been elected slot leader.
 - Providing necessary network infrastructure.

Incentive Types



- In this talk, when we talk about incentives, we mean **monetary** incentives in the form of ADA.
- There are other types of incentives as well: things like **idealism** and **morality** and the general desire to “do the right thing”.
- Design goal for Cardano incentives: **Monetary and moral incentives should align perfectly.**

Danger of Misalignment



- In 2014, the mining pool GHash.io controlled more than 51% of the Bitcoin mining power for several days.
- Eventually, GHash.io voluntarily reduced its mining power and promised not to exceed 40% in the future.
- There was a real conflict between the financial interests of the miners and the security of the network.

Desired Configuration



- A solid majority of stake (ca. 80%) should be delegated to a number of k stake pools ($k \sim 100$ seems to be reasonable).
- The stake pools should be online when needed, and they should provide additional network infrastructure ("relay nodes").

Incentive Sources



- Transaction fees.
- Non-refundable deposits.
- Monetary expansion.

Incentives Distribution



- In Cardano, time is divided into **epochs** and **slots**.
- A slot lasts 20 seconds, an epoch contains 21,600 slots and lasts five days.
- Incentives are distributed on an **epoch by epoch** base:
Transaction fees, deposits and monetary expansion are collected into a virtual rewards pool; then this pool is distributed amongst the stakeholders.

Basic Idea of Distribution



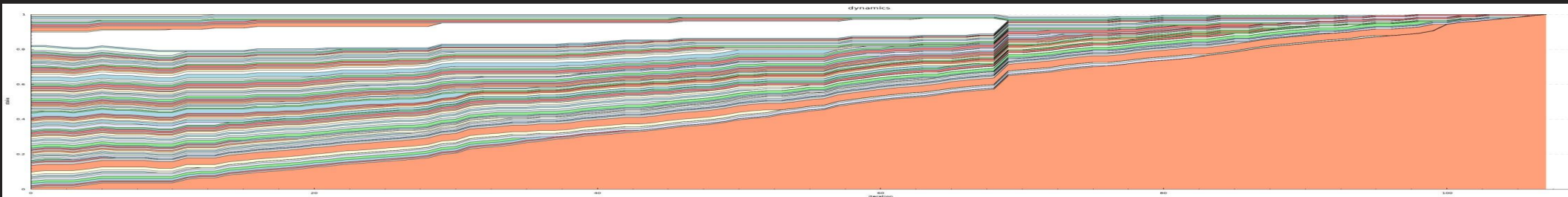
The rewards pool from one epoch is distributed amongst stake pools (and individual protocol participants) **according to their stake.**

Problem with the Basic Idea



The basic idea is a good guideline, but too naive: The fewer pools there are, the lower total costs will be, the higher everybody's rewards will be.

So the system will tend towards a single dictatorial pool that everybody else delegates to.



First refinement: Large Pools



The maximal proportion of the rewards pool that a stake pool can receive will be limited by $1/k$, where k is the number of desired pools ($k \sim 100$).

Second refinement: Being Online

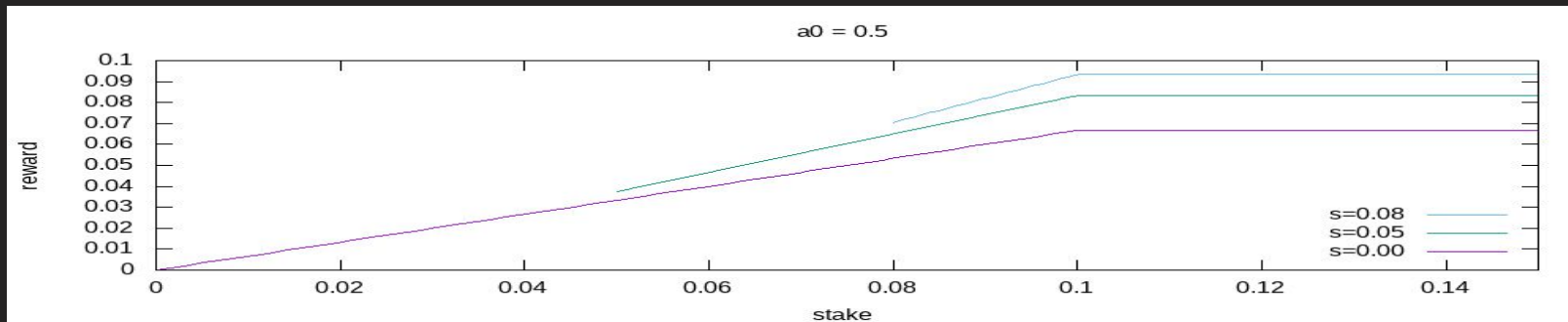


- Stake pools should be **penalized** for **not** following the protocol and **not** being online when it is their turn.
- Rewards will be proportional to **performance**.
- In a protocol without public leader schedule like Ouroboros Praos, performance has to be **estimated**.

Third refinement: Sybil Prevention



- An attacker could create hundreds of "attractive" pools and have more than 50% of people delegating to one of them.
- Handled by making pool operators "pledge" some stake to their pools and make pool rewards depend on the pledged amount.



Undistributed Rewards



- These refinements can lead to a situation where not all funds contained in the rewards pool will be distributed.
- This, however, is a feature, not a bug, because the remaining funds can instead be put to use in the **treasury**.

Distribution to Pool Members



- The pool **leader** herself should be compensated for her costs (computing power, online time) and rewarded for her efforts.
- Pool **members** should be rewarded proportional to the stake they delegated to the pool.

Not Being Short Sighted



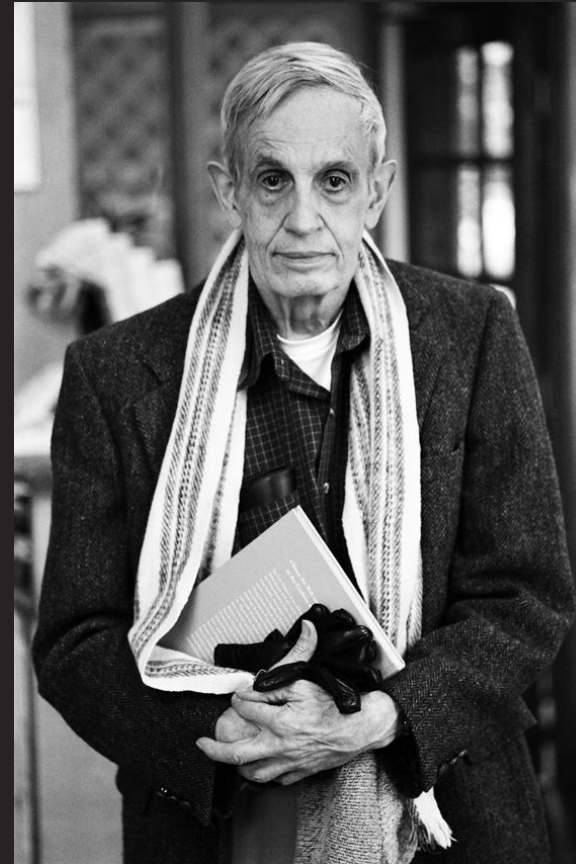
- It might seem profitable for a pool operator to change his strategy and increase his margin.
- In reality, of course, pool operators will know that people will leave their pools if they do that.
- So expected rewards displayed in the wallet will “look ahead” and take into account that only the k most attractive pools will actually have members.

Game Theory



.John von Neumann
(Los Alamos).

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.John Forbes Nash Jr.
(June 13, 1928 – May 23, 2015)

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Nash Equilibria



- In game theory, Nash's concept of a **Nash Equilibrium** is of central importance.
- A Nash Equilibrium is a situation where no player can improve his payoff by unilaterally changing his strategy.
- So if all other players stick to their strategies, it would be irrational for any player to change his own strategy.

Prisoner's Dilemma



	Burglar B confesses	Burglar B stays silent
Burglar A confesses	Burglar A: 5 years, Burglar B: 5 years	Burglar A goes free, Burglar B: 10 years
Burglar A stays silent	Burglar A: 10 years, Burglar B: goes free	Burglar A: 1 year, Burglar B: 1 year

This game has exactly one **Nash Equilibrium**: Both burglars confess and spend five years in prison.

The Staking Game



- In a 2022 research paper, game theory was applied to the Cardano incentives mechanism.
- It was shown that if the reward distribution follows the rules explained above, then each Nash Equilibrium of the "staking game" will have k stake pools of the same size.



Thank you!